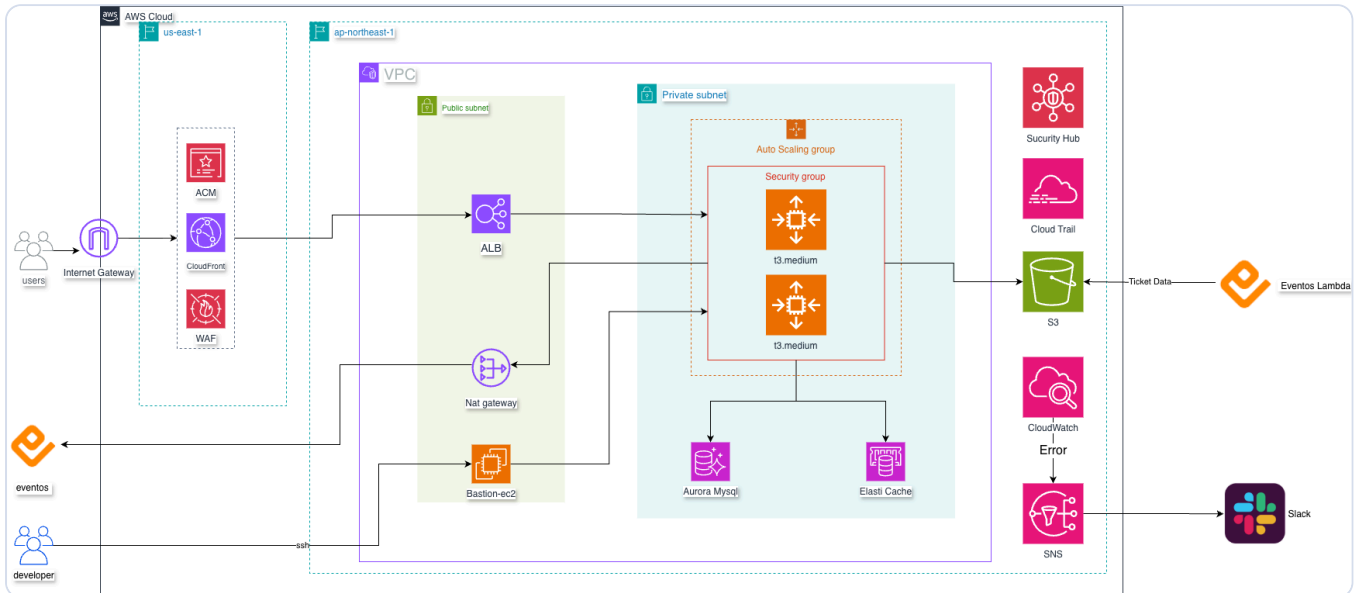


AWS Console Setup Runbook

Phiên bản đơn giản cho môi trường test: chỉ tạo VPC + EC2 (không có ALB, Aurora, Redis, NAT Gateway). Chi phí gần như \$0 nếu dùng t3.micro trong free tier.

Sơ đồ kiến trúc tổng quan hệ thống:



Phase 1.1 — Tạo VPC

- Name: `eventtech-dev-vpc`
- CIDR: `10.40.0.0/16`
- Bật **DNS resolution + DNS hostnames**

MỤC ĐÍCH

Mạng riêng ảo biệt lập cho môi trường test.

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional Info
Creates a tag with a key of 'Name' and a value that you specify.

my-vpc-01

IPv4 CIDR block Info
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/24
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy Info
Default

VPC encryption control (\$) Info
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode

Monitor mode: See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.

Enforce mode: Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Phase 1.2 — Tạo Subnets

- 1 public subnet: `eventech-dev-public-1a`
- AZ: `ap-northeast-1a`
- CIDR: `10.40.0.0/24`
- **Bật auto-assign public IPv4**

MỤC ĐÍCH

Subnet public để EC2 có thể truy cập internet trực tiếp (test environment không cần private subnet).

The screenshot shows the 'Create internet gateway' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Internet gateways > Create internet gateway'. The page title is 'Create internet gateway' with an 'Info' link. A description states: 'An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.' The 'Internet gateway settings' section includes a 'Name tag' field with the value 'my-internet-gateway'. Below this is a 'Tags - optional' section with a description of tags and a note that no tags are currently associated with the resource. An 'Add new tag' button is present. At the bottom right are 'Cancel' and 'Create internet gateway' buttons. The footer includes links for CloudShell, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services, Inc.

Phase 1.3 — Tạo Internet Gateway

- Name: `eventech-dev-igw`
- Attach vào `eventech-dev-vpc`

MỤC ĐÍCH

Kết nối VPC với internet.

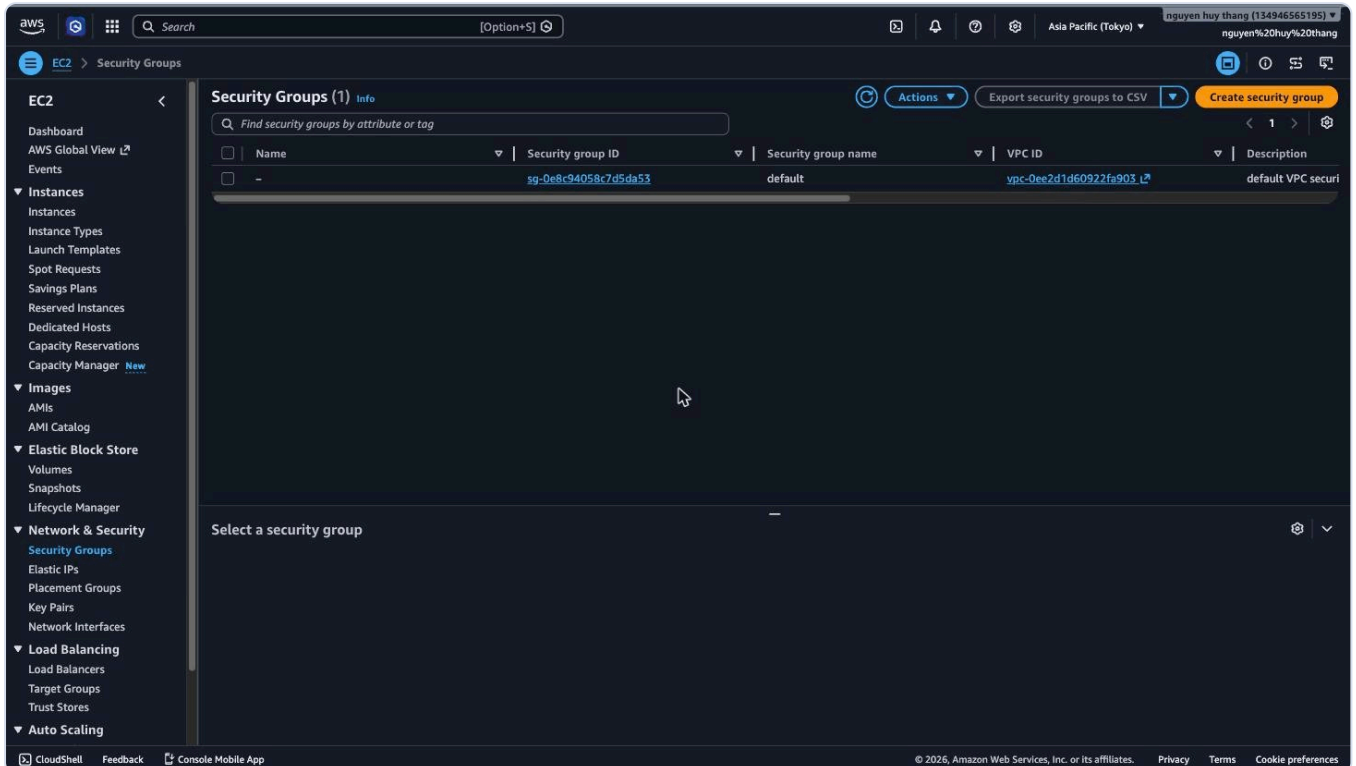
The screenshot shows the 'Create route table' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Route tables > Create route table'. The page title is 'Create route table' with an 'Info' link. A description states: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.' The 'Route table settings' section includes a 'Name - optional' field with the value 'my-route-table-01'. Below this is a 'VPC' dropdown menu with the text 'Select a VPC'. At the bottom right are 'Cancel' and 'Create route table' buttons. The footer includes links for CloudShell, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services, Inc.

Phase 1.4 — Tạo Route Table

- Name: `eventech-dev-public-rt`
- Add route: `0.0.0.0/0` → `eventech-dev-igw`
- Associate: `eventech-dev-public-1a`

MỤC ĐÍCH

Cho phép traffic từ subnet đi ra internet.



Phase 1.5 — Tạo Security Group

- Name: `eventech-dev-ec2-sg`
- **Inbound rules:**
 - SSH `22` from `YOUR_PUBLIC_IP/32`
 - HTTP `80` from `0.0.0.0/0`
 - HTTPS `443` from `0.0.0.0/0`
 - Custom TCP `8080` from `0.0.0.0/0`
- **Outbound:** All traffic

MỤC ĐÍCH

Firewall kiểm soát traffic vào EC2, chỉ cho phép SSH từ IP của bạn để bảo mật.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name
 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Instance type Info | Get advice

Instance type
 ☐ All generations [Compare instance types](#)

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*
 [Create new key pair](#)

Network settings Info [Edit](#)

Network Info

Summary

Number of instances Info

Software Image (AMI)
 -

Virtual server type (instance type)
 -

Firewall (security group)
 -

Storage (volumes)
 -

[Cancel](#) [Launch instance](#) [Preview code](#)

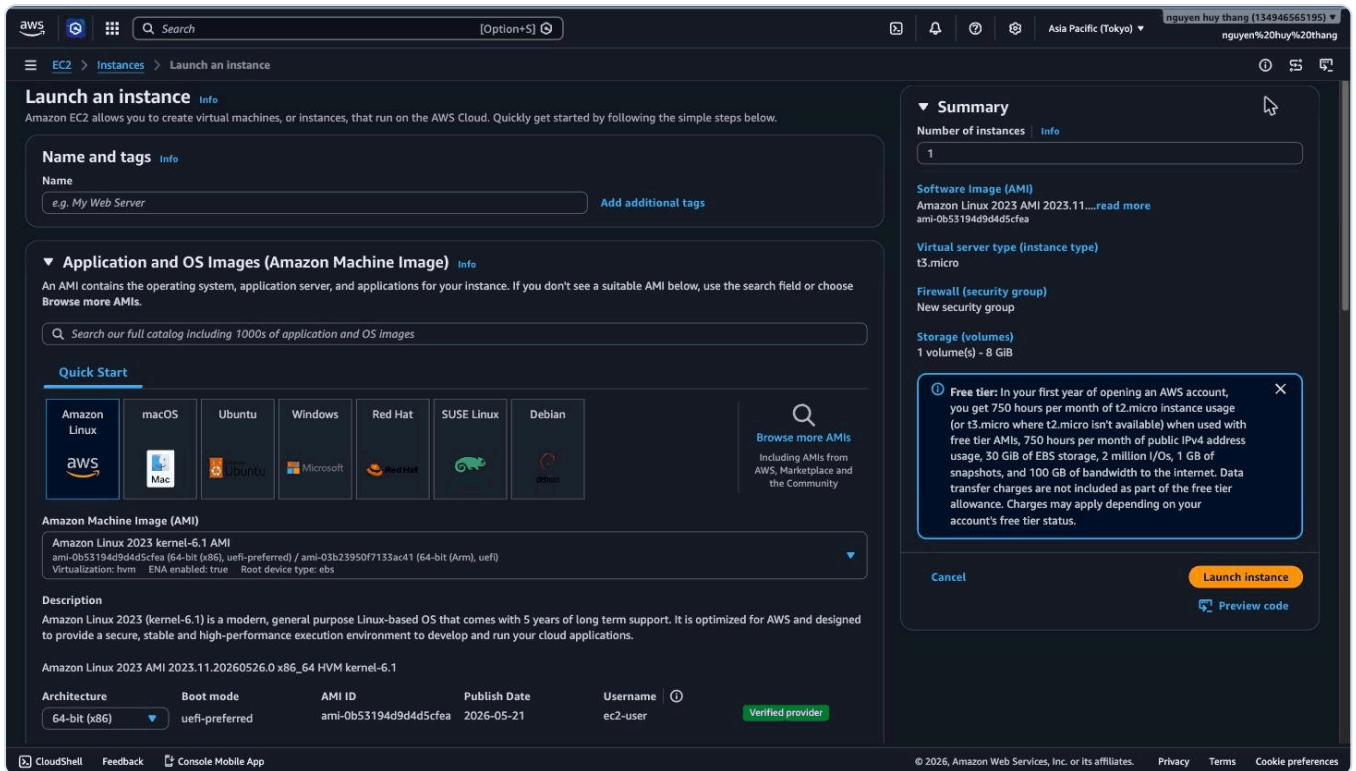
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Phase 1.6 — Tạo EC2 Instance

- Name: `eventech-dev-app`
- AMI: **Amazon Linux 2023**
- Instance type: `t3.micro`
- Key pair: tạo mới hoặc chọn key pair có sẵn
- Network: `eventech-dev-vpc`
- Subnet: `eventech-dev-public-1a`
- Auto-assign public IP: **Enable**
- Security group: `eventech-dev-ec2-sg`
- Storage: **20 GiB gp3**

MỤC ĐÍCH

Server chạy ứng dụng, dùng t3.micro (free tier eligible) cho môi trường test.



Phase 1.7 — Kết nối SSH vào EC2

BASH

```
# Lấy Public IP từ EC2 Console
ssh -i your-key.pem ec2-user@<public-ip>

# Nếu dùng Ubuntu AMI
ssh -i your-key.pem ubuntu@<public-ip>
```

MỤC ĐÍCH

Truy cập vào server để cài đặt và cấu hình ứng dụng.

Phase 1.8 — Cài đặt ứng dụng trên EC2

BASH

```
# Cập nhật packages
sudo yum update -y # Amazon Linux
# hoặc
sudo apt update && sudo apt upgrade -y # Ubuntu

# Cài Nginx
sudo yum install nginx -y # Amazon Linux
sudo systemctl start nginx
sudo systemctl enable nginx

# Kiểm tra
curl http://localhost
```

MỤC ĐÍCH

Cài web server để test ứng dụng.

Phase 1.9 — Test kết nối

TEXT

Lấy Public IPv4 DNS từ EC2 Console:
`http://<public-ip>`
`http://<public-ipv4-dns>`

Kiểm tra:

- Public IP hiển thị trong EC2 Console
- Truy cập `http://<public-ip>` từ browser
- SSH kết nối được

Dọn dẹp sau khi test xong (tránh phát sinh phí)

TEXT

```
[ ] Terminate EC2 instance
[ ] Delete Security Group: eventech-dev-ec2-sg
[ ] Detach và Delete Internet Gateway: eventech-dev-igw
[ ] Delete Subnet: eventech-dev-public-1a
[ ] Delete Route Table: eventech-dev-public-rt
[ ] Delete VPC: eventech-dev-vpc
```

Thứ tự xoá quan trọng — phải terminate EC2 trước, sau đó mới xoá Security Group, IGW, Subnet, Route Table, VPC.

Generated from AWS_CONSOLE_SETUP_RUNBOOK.md — Simplified for test environment

Phần 2 — Mở rộng: ALB, NAT Gateway, Bastion

⚠ LƯU Ý CHI PHÍ

NAT Gateway: ~\$45/tháng | ALB: ~\$17/tháng | Bastion t3.micro: ~\$7.5/tháng (free tier năm đầu = \$0)
XÓA NGAY SAU KHI TEST XONG ĐỂ TRÁNH TÍNH PHÍ!

Phase 2.1 — Thêm Public Subnet 1c (cần cho ALB)

MỤC ĐÍCH

ALB yêu cầu ít nhất 2 Availability Zone. Cần thêm subnet ở AZ thứ hai (ap-northeast-1c).

Field	Giá trị
Subnet name	eventech-dev-public-1c
Availability Zone	ap-northeast-1c
IPv4 CIDR block	10.40.1.0/24
VPC	eventech-dev-vpc

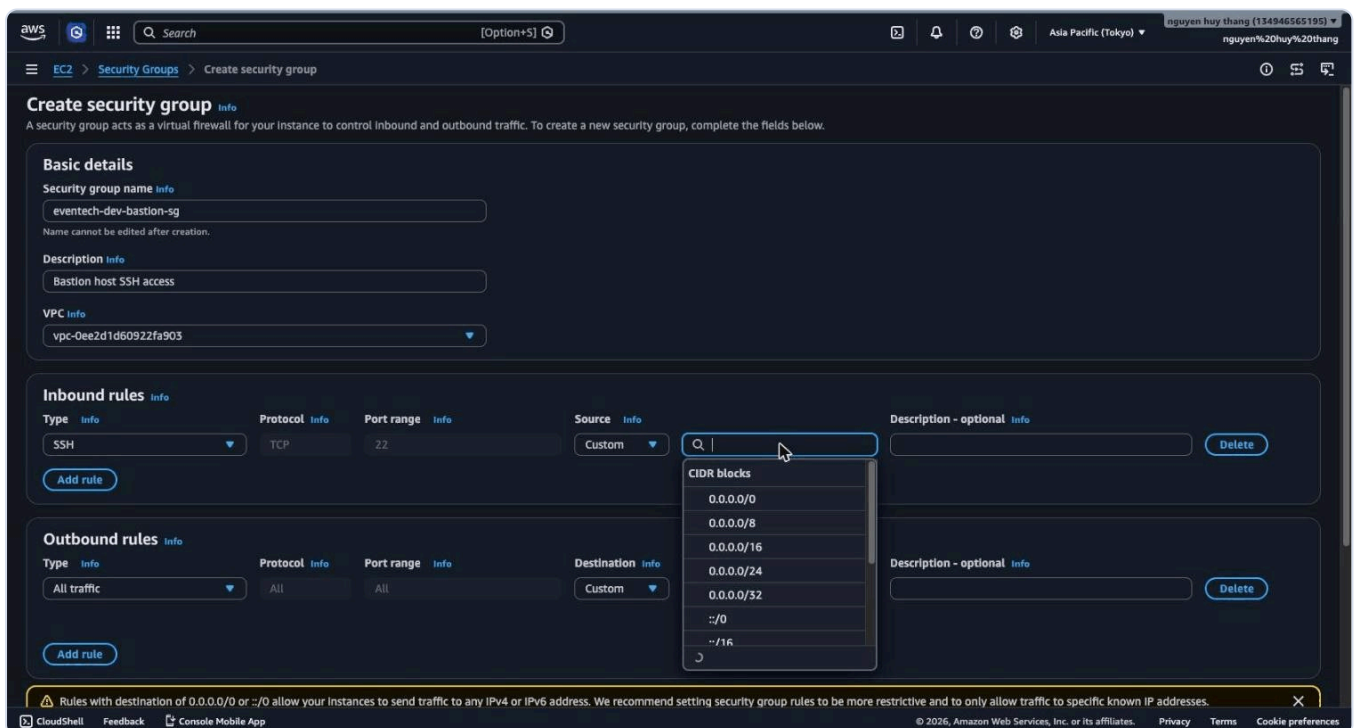
Vào **VPC** → **Subnets** → **Create subnet**, chọn VPC eventech-dev-vpc, điền thông tin trên.

Phase 2.2 — Tạo Security Group Bastion (SSH Access)

MỤC ĐÍCH

Kiểm soát SSH (port 22) vào Bastion host, chỉ cho phép từ IP của bạn.

Field	Giá trị
Security group name	eventech-dev-bastion-sg
Description	Bastion host SSH access
VPC	eventech-dev-vpc
Inbound rule — Type	SSH
Inbound rule — Port	22
Inbound rule — Source	YOUR_PUBLIC_IP/32 (chạy: <code>curl ifconfig.me</code>)

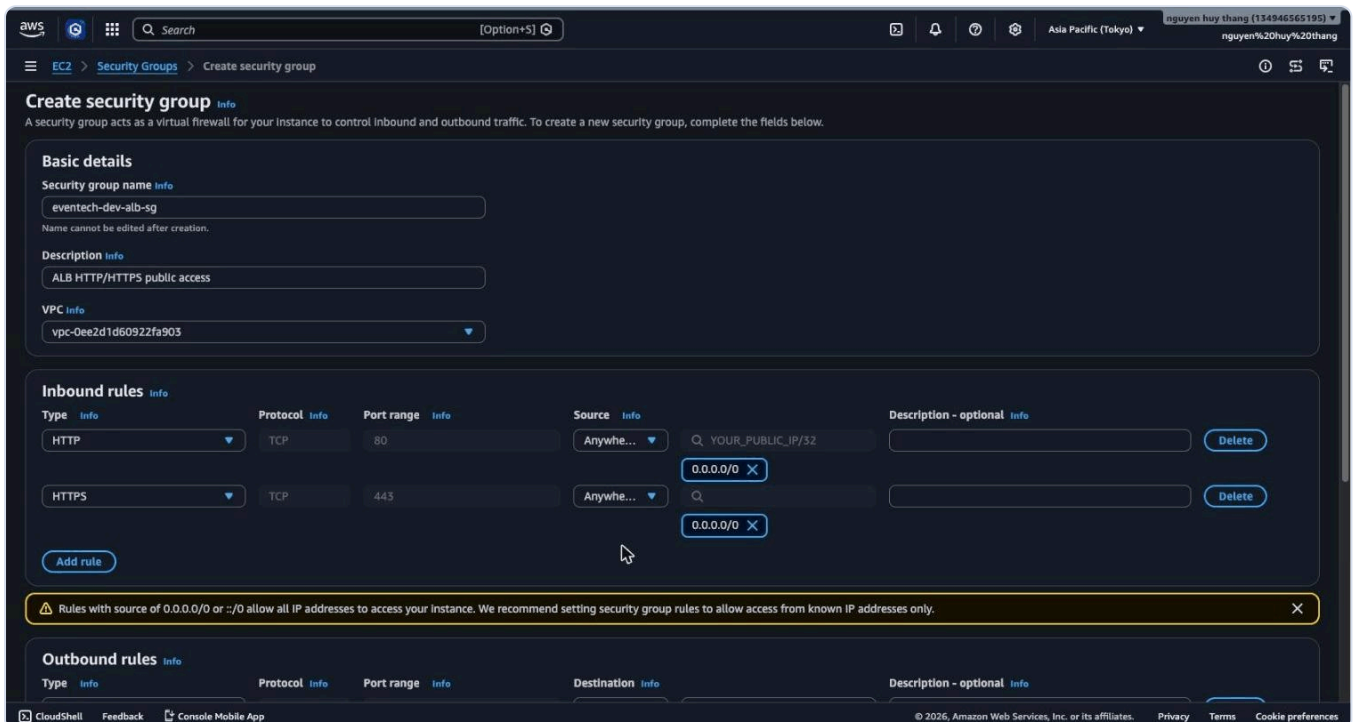


Phase 2.3 — Tạo Security Group ALB (HTTP/HTTPS)

MỤC ĐÍCH

Cho phép traffic HTTP (80) và HTTPS (443) từ Internet vào Application Load Balancer.

Field	Giá trị
Security group name	eventech-dev-alb-sg
Description	ALB HTTP/HTTPS public access
VPC	eventech-dev-vpc
Inbound rule 1	HTTP, Port 80, Source: 0.0.0.0/0
Inbound rule 2	HTTPS, Port 443, Source: 0.0.0.0/0



Phase 2.4 — Tạo NAT Gateway ⚠ ~\$45/tháng

⚠ CẢNH BÁO CHI PHÍ

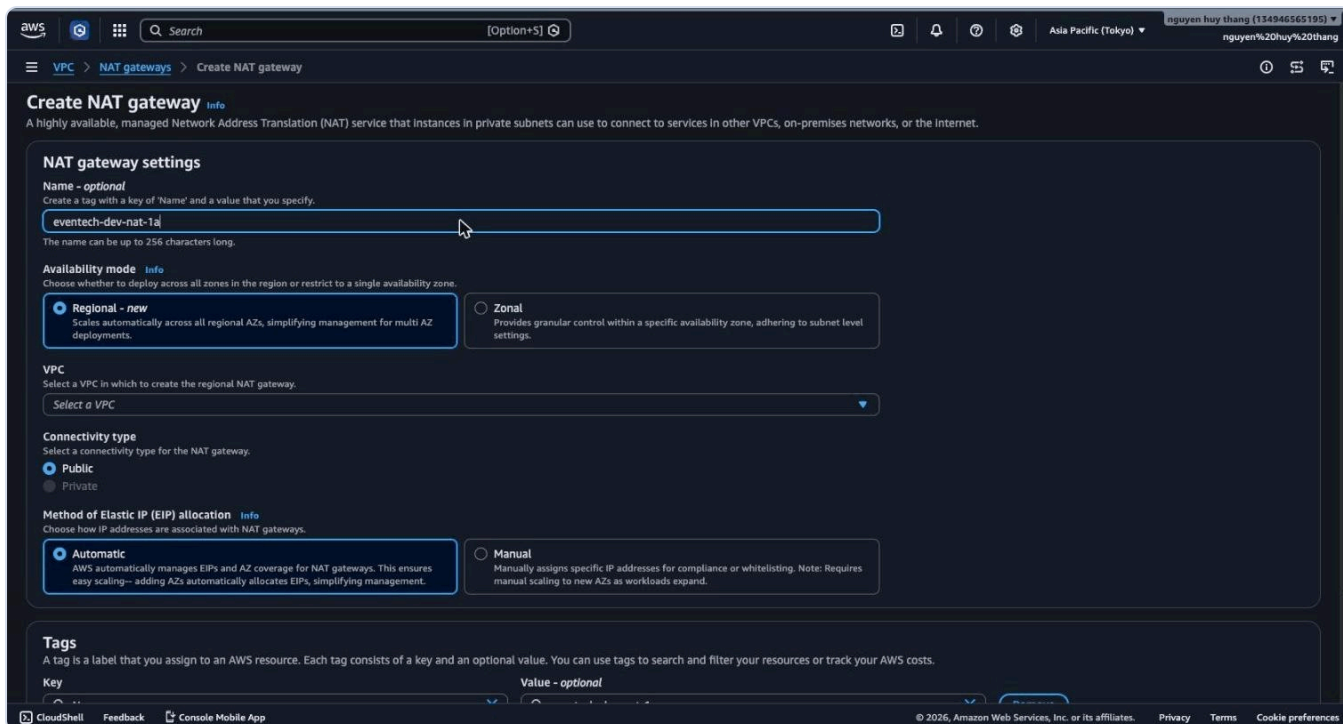
NAT Gateway tính phí kể cả khi idle! Xóa ngay sau khi không dùng nữa.

MỤC ĐÍCH

Cho phép EC2 trong private subnet kết nối ra Internet (download packages, updates) mà không cần public IP.

Field	Giá trị
Name	eventech-dev-nat-1a
Subnet	eventech-dev-public-1a (public subnet)
Connectivity type	Public
Elastic IP	Click Allocate Elastic IP

VPC → NAT gateways → Create NAT gateway. Chờ Status = **Available** (vài phút).



Phase 2.5 — Tạo Bastion EC2

MỤC ĐÍCH

Jump server để SSH vào các EC2 trong private subnet, tăng bảo mật bằng cách không expose private EC2 trực tiếp ra Internet.

Field	Giá trị
Name	eventech-dev-bastion
AMI	Amazon Linux 2023
Instance type	t3.micro
Key pair	Chọn hoặc tạo key pair
Subnet	eventech-dev-public-1a
Auto-assign public IP	Enable
Security group	eventech-dev-bastion-sg

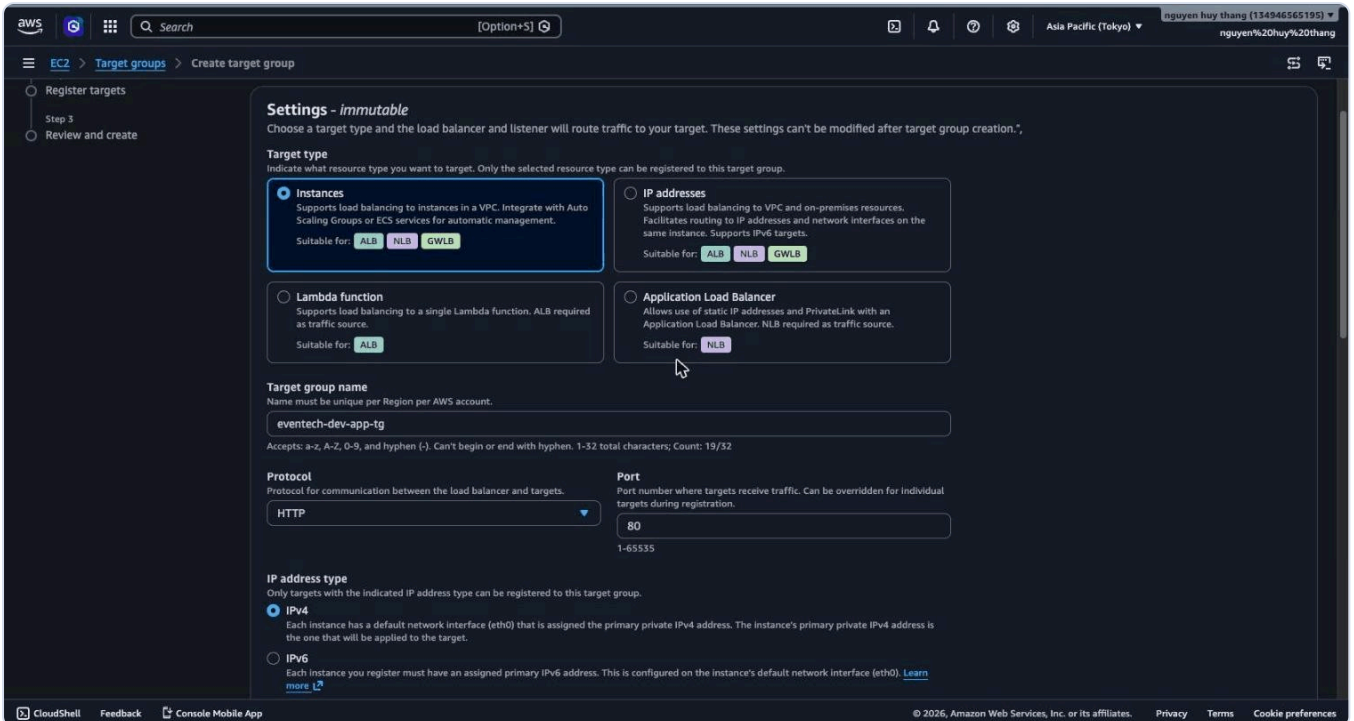
EC2 → Launch instances → điền các thông tin trên → Launch instance.

Phase 2.6 — Tạo Target Group

MỤC ĐÍCH

Nhóm các EC2 instances sẽ nhận traffic từ ALB. Health check đảm bảo chỉ instance healthy mới nhận request.

Field	Giá trị
Target type	Instances

Target group name `eventech-dev-app-tg`Protocol `HTTP`Port `80`VPC `eventech-dev-vpc`Health check path `/`

Phase 2.7 — Tạo Application Load Balancer (ALB) ⚠️ ~\$17/tháng

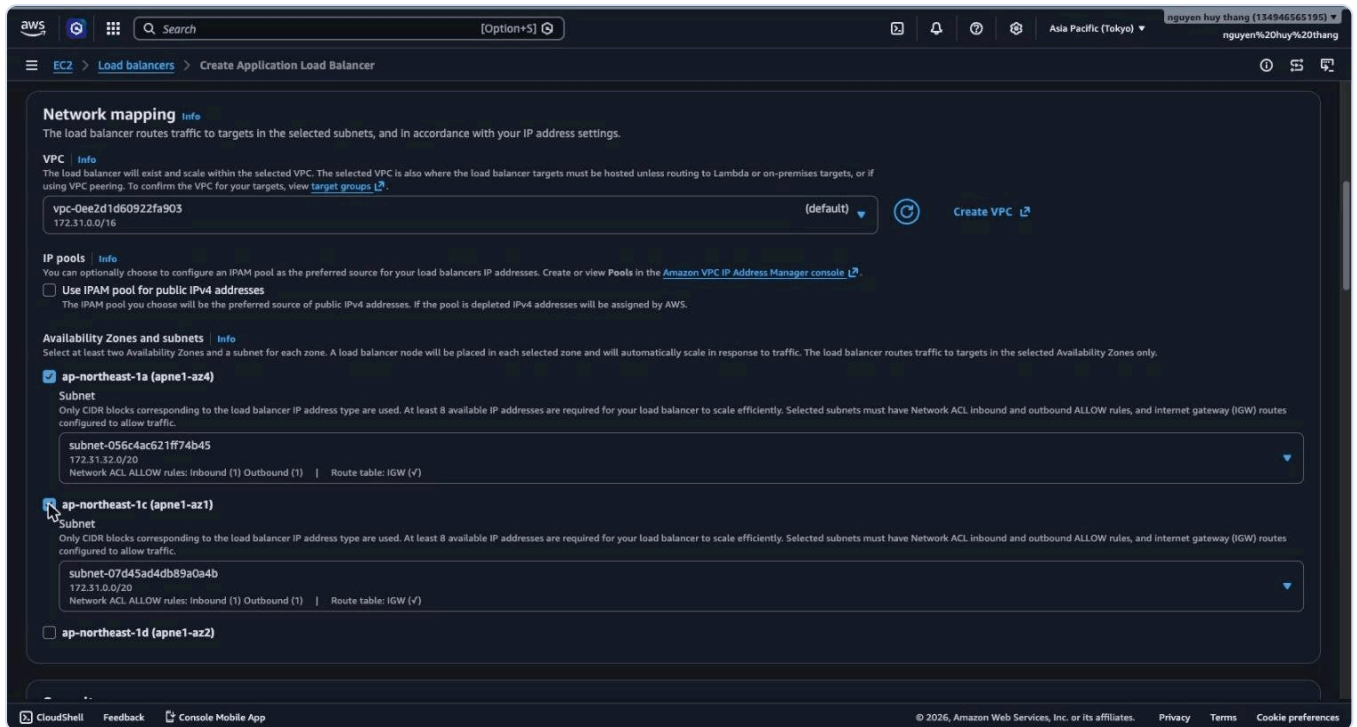
⚠️ CẢNH BÁO CHI PHÍ

ALB tính phí liên tục. Xóa ngay sau khi không dùng nữa.

MỤC ĐÍCH

Phân phối traffic HTTP/HTTPS từ Internet đến các EC2 instances, hỗ trợ health check, SSL termination, và routing rules.

Field	Giá trị
Load balancer name	<code>eventech-dev-alb</code>
Scheme	<code>Internet-facing</code>
IP address type	<code>IPv4</code>
VPC	<code>eventech-dev-vpc</code>
Availability Zones	<code>ap-northeast-1a</code> + <code>ap-northeast-1c</code> (chọn cả 2)
Security group	<code>eventech-dev-alb-sg</code>
Listener	HTTP:80 → Forward to <code>eventech-dev-app-tg</code>



Cleanup — Thứ tự xóa tài nguyên (quan trọng!)

XÓA ĐÚNG THỨ TỰ ĐỂ TRÁNH DEPENDENCY ERROR:

1. Delete ALB (EC2 → Load Balancers)
2. Delete Target Group (EC2 → Target Groups)
3. Terminate Bastion EC2
4. Delete NAT Gateway → Release Elastic IP
(NAT GW TÍNH PHÍ KỂ CẢ IDLE!)
5. Delete Security Groups: alb-sg, bastion-sg
6. Delete Subnets → Route Tables → IGW → VPC